

HW#2

Due: February 17th, 2022

CBE450: Process Design - Spring 2022

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Supply Chain for Plastic Waste Processing (100 pts)

Plastic waste affects the environment in many ways; it is estimated that 3.5 million tons of plastic waste are generated every year in the Upper Midwest region of the United States. Fortunately, there is a new promising approach for dealing with plastic waste in a scalable manner. This pathway uses thermochemical technologies (pyrolysis and steam cracking) to break plastic and recover value-added chemicals. As shown in Figure 1, pyrolysis converts the plastic waste into a pyrolysis oil, while steam cracking produces ethylene from the pyrolysis oil.

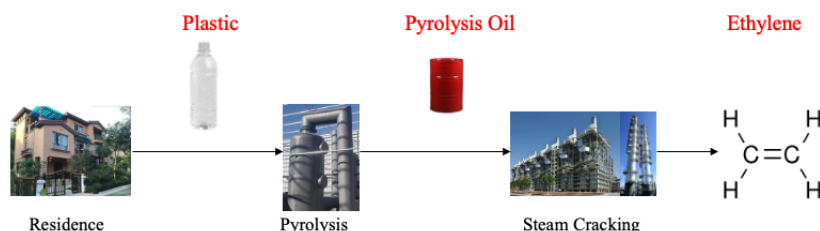


Figure 1: Schematic of plastic waste processing pathway

We would like to design and operate a supply chain that collects and processes waste and obtains value-added products (ethylene). You are given the following data:

- **Supply Information:** The goal is to recycle all the plastic waste generated in Madison (MAD) and Milwaukee (MKE). The total plastic waste generated in MAD and MKE is 33100, 77300 ton/yr, respectively. The supply cost for the plastic waste from these cities is 0 \$/ton (plastic waste is offered for free).
- **Demand Information:** Each city has a consumer of ethylene and we wish to satisfy the demands of such consumers. The demand capacities for ethylene are both 100,000 tons/yr and the consumers are both willing to pay 1050 \$/ton.
- **Transportation Information:** The transportation cost for the three products (plastic, pyrolysis oil, and ethylene) are the same. The cost to move one ton of product from MAD to MKE (and from MKE to MAD) is 10\$/ton. The transportation capacity for all products is 100,000 ton/yr.
- **Technology Information:**
 - You have the possibility of installing a pyrolysis process in MAD and another one in MKE; you also have the possibility of installing a steam cracking process in MAD and another one in MKE.
 - For the pyrolysis process, 0.7 ton of pyrolysis oil can be produced from breaking one ton of plastic waste. The operational cost of pyrolyzing one ton of plastic waste is 14 \$. The maximum capacity of a single pyrolysis facility is 100,000 tons of plastic waste per year. The investment cost for this technology is 2,000,000 \$/yr (annualized investment).
 - The steam cracking process converts one ton of pyrolysis oil into 0.25 ton of ethylene. The operational cost of steam cracking is 71\$ per ton of pyrolysis oil. A single steam cracking facility process 100,000 ton of pyrolysis oil each year. The investment cost for this technology is 2,800,000 \$/yr (annualized investment).

Use **Julia** to perform the following tasks:

- (10 pts) Identify the sets of all participants in the supply chain.
- (10 pts) Identify all the data in the supply chain.
- (10 pts) Draw a diagram of the supply chain.
- (10 pts) Solve a supply chain design problem to identify the optimal technology selection (and location) and product allocations that maximize the value of demand served and that minimize the cost of suppliers, transport, and processing (operation and investment).
- (10 pts) Display the technology selection and optimal allocations in the diagram of the supply chain. Confirm that the allocations satisfy all balances at all locations and for all products.
- (10 pts) Fix the design obtained (technology selection) and solve a supply chain management problem to calculate the prices for all products and the costs, revenues, and profits for all the participants in the supply chain.
- (10 pts) Calculate the total profit of the supply chain.
- (10 pts) What is the actual value (price) of plastic waste? Is plastic waste in fact a valuable product? What profit do suppliers make by selling their waste?
- (10 pts) Does the supply chain designed recycles all the plastic waste available? Analyze the solution to explain why not.
- (10 pts) What is the minimum amount that consumers should be willing to pay for ethylene so that we can obtain a supply chain design that processes *all* the available plastic waste? What is the actual value (price) of plastic waste under this new supply chain design?